3.2.4 - 3.3.5

Wednesday, April 8, 2026 7:35 PM

Exercise 12. Consider exercise 8 again. Which pairs (a,b) from that exercise are coprime?

8 (45,10)

k	r	s	t	q
0	45	1	0	
1	10	0	1	
2	5	1	-4	4
	0	2	9	2

gcd = 5
s = 1
t = -4

$5 = 1 \cdot 45 - 4 \cdot 10$

(13,11)

	13	1	0	
	11	0	1	
2	2	1	-1	1
	1	-5	6	5
	0	11	-13	2

gcd = 1
s = -5
t = 6

$1 = -5 \cdot 13 + 6 \cdot 11$

(13,12)

13	1	0
12	0	1
1	1	-1
0		12

gcd = 1
s = 1
t = -1

$1 = 1 \cdot 13 + -1 \cdot 12$

12 (13,11)
(13,12)

Exercise 13. Consider the **octal** positional system, which represents integers with 8 digits, usually written as $\{0,1,2,3,4,5,6,7\}$. Numbers in this system are characterized by the prefix $0o$. Write the numbers $0o1354$ and $0o777$ into their decimal representation.

$0o1354 = 748$

$8^0 \cdot 4 + 8^1 \cdot 5 + 8^2 \cdot 3 + 8^3 \cdot 1$

$1 \cdot 4 + 40 + 64 \cdot 3 + 512$

$4 + 40 + 192 + 512$

$$\begin{array}{r} 192 \\ 40 \\ 4 \\ \hline 748 \end{array} +$$

$0o777$

$8^0 \cdot 7 + 8^1 \cdot 7 + 8^2 \cdot 7$

$1 \cdot 7 + 56 + 64 \cdot 7$

$8^0 \cdot 7 + 8^1 \cdot 7 + 8^2 \cdot 7$

$1 \cdot 7 + 56 + 64 \cdot 7$

$7 + 56 + 448$

518

Exercise 14. Which of the following pairs of numbers are congruent with respect to the modulus 13:

- $(5,19)$ ✗
- $(13,0)$ ✓
- $(-4,9)$ ✓
- $(0,0)$ ✓

Exercise 15. Find all integers x, such that the congruence $x \equiv 4 \pmod 6$ is satisfied.

$6k+4$

$\{\ldots -2, 4, 10 \ldots\}$

Exercise 16. Consider the modulus 13 and find all solutions $x \in \mathbb{Z}$ to the following congruence:

$$5x+4 \equiv 28+2x \pmod{13}$$

Exercise 17. Consider the modulus 23 and find all solutions $x \in \mathbb{Z}$ to the following congruence:

$$69x \equiv 5 \pmod{23}$$

Exercise 18. Consider the modulus 23 and find all solutions $x \in \mathbb{Z}$ to the following congruence:

$$69x \equiv 46 \pmod{23}$$

Exercise 19. Let a,b,k be integers, such that $a \equiv b \pmod n$ holds. Show $a^k \equiv b^k \pmod n$.

Exercise 20. Let a,n be integers, such that a and n are not coprime. For which $b \in \mathbb{Z}$ does the congruence $a \cdot x \equiv b \pmod n$ have a solution x and how does the solution set look in that case?

16 $5x+4 \equiv 28+2x \pmod{13}$

$3x \equiv 24 \mod 13$

$\gcd(3,13)=1$

$x \equiv 8 \mod 13$

$8+13k \quad k \in \mathbb{Z}$

17 $69x \equiv 5 \mod 23$

$0.x \equiv 5 \mod 23$

$\emptyset$

18 $69x \equiv 46 \mod 23$

$0.x \equiv 0 \mod 23$

$x \in \mathbb{Z}$

Exercise 21. Define $\mathbb{Z}_{13}$ as the the arithmetic modulo 13 analogously to example 11. Then consider the congruence from exercise 16 and rewrite it into an equation in $\mathbb{Z}_{13}$.

$5x+4 \equiv 28+2x \mod 13$

$5x+4 = 2+2x$ over $\mathbb{Z}_{13}$

Exercise 22. Consider the modulus $n = 24$. Which of the integers 7, 1, 0, 805, -4255 have multiplicative inverses in modular 24 arithmetic? Compute the inverses, in case they exist.

Exercise 23. Find the set of all solutions to the congruence $17(2x+5)-4 \equiv 2x+4 \pmod 5$. Then project the congruence into $\mathbb{Z}_5$ and solve the resulting equation in $\mathbb{Z}_5$. Compare the results.

Exercise 24. Find the set of all solutions to the congruence $17(2x+5)-4 \equiv 2x+4 \pmod 6$. Then project the congruence into $\mathbb{Z}_6$ and try to solve the resulting equation in $\mathbb{Z}_6$.

22)

$\to 7^{-1}$ in $\mathbb{Z}_{24}$ $= 7$

$\to 1^{-1}$ $= 1$

$\to 0^{-1}$ $\emptyset$

$805 \equiv 13$

k	r	s	t	q
	24	1	0	
	13	0	1	
	11	1	-1	1
	2	-1	2	1
	1	6	-11	5
	0			2

$1 = 24.6 - 11.13$

$\to 805^{-1} = -11$

$-4255 \equiv 17$

k	r	s	t	q
	24	1	0	
	17	0	1	
	7	1	-1	1
	3	-2	3	2
	1	5	-7	2
	0			3

$1 = 24 . 5 - 7 . 17$

$\to -4255^{-1} = -7$ in $\mathbb{Z}_{24}$

23) $17(2x+5)-4 \equiv 2x+4 \mod 5$ $\{-1, 4, 9, \ldots\}$

projecting

23) $17(2x+5)-4 \equiv 2x+4 \mod 5$ $\{\ldots, 1, 4, 9, \ldots\}$

projecting

$2(2x)-4 = 2x+4$ over $\mathbb{Z}_5$

$4x-2x = 4+4$

$2x = 8$

$x = 4$

24) $17(2x+5)-4 \equiv 2x+4 \mod 6$

$5(2x+5)+2 = 2x+4$ over $\mathbb{Z}_6$

$4x+1+2 = 2x+4$

$2x = 1$

no solution